Parametrization of Peak Prediction Using Quantile Regression and Asymmetric Laplace Noise

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Abstract

In many signal processing applications, defining and finding peaks is an important part of the pipeline. Peak prediction can be a very challenging endeavor, especially when there is a lot of noise. In this work, the peak region is considered as a combination of a Gaussian trend with additive asymmetric Laplace noise, and the peak is modeled as the peak of the Gaussian trend.

We propose a criterion based on the weighted quantile which predicts the optimal number of observations necessary for the exact prediction of the peak's location. We analyze the effect of the signal-to-noise ratio on our prediction. We show that for small signal-to-noise ratio (a lot of noise), the prediction becomes inaccurate.

In this case, we propose a new idea called the entry time into the region of the peak and the exit time from the region of the peak. We demonstrate how to apply our new idea to real data and we analyze the effect of the choice of the objective function. As a numerical application, we consider the peak prediction of the daily infections during the first wave of COVID-19 in China, France, and Germany.

Keywords: Gaussian model, least absolute deviation, weighted median, entry time, region of the peak, daily infection.

Mathematics subject classification: 62M10, 60G35.

1 Introduction

Predicting peaks in time series data is crucial across various domains for proactive decision-making, risk management, resource allocation, and disaster preparedness. The ability to anticipate spikes in data can serve as an early warning system for events such as disease outbreaks in epidemiology, allowing public health officials to allocate resources and implement preventive measures effectively. In energy management, predicting peak electricity demand enables utility companies to adjust generation and distribution schedules, ensuring a reliable power supply

and preventing shortages. Financial analysts use peak prediction to manage investment risks by anticipating fluctuations in stock prices or market volatility. Additionally, environmental scientists rely on peak prediction to monitor pollution levels and natural disasters, facilitating disaster preparedness and response efforts. The examples justifying the usefulness of locating, measuring, and understanding these peaks are numerous.

Predicting peaks in time series data is a complex task due to several reasons. Firstly, the presence of noise and variability in the data makes it difficult to accurately identify and predict peaks. This noise can originate from various sources, such as measurement errors or random fluctuations in the data. Additionally, many real-world systems exhibit nonlinear dynamics, meaning that the relationship between variables is not strictly linear. Nonlinearities can lead to unexpected behaviors in the data, further complicating peak prediction. Moreover, time series data often contain complex patterns, such as multiple overlapping peaks or irregular intervals between peaks, making it challenging to develop accurate prediction models. In the literature, existing peak prediction methods often rely on specific assumptions about the data distribution or peak characteristics, which may not always hold true in practice, leading to inaccurate predictions (see for instance, **?** and **?**).

Popular forecasting models such as Autoregressive Integrated Moving Average (ARIMA) and Recurrent Neural Networks (RNN) often struggle to accurately define or forecast peak values in time series data. Despite their effectiveness in capturing general trends and patterns, these models may overlook or underestimate the occurrence of peaks due to their complexity and the inherent variability of peak events.

In the realm of peak prediction, there has been relatively little theoretical work compared to more generalized forecasting approaches. One notable contribution in this area is Tucker thesis **?**, which delved into the dynamics of trend, seasonality, and noise in time series data to predict peaks. Tucker research explored how the interplay between these components could be leveraged to identify and forecast peak events accurately. Inspired by previous studies **?**, Tucker applied a combination of general prediction techniques and specialized peak prediction methods. He utilized the Peaks function available in Scipy's Signal module, which offers tools for detecting and analyzing peak values in data. Tucker thesis represents a significant step forward in the field of peak prediction, offering insights into the complex dynamics underlying peak events and proposing practical methods for their prediction. By building on existing theoretical frameworks and leveraging advanced computational tools, Tucker work contributes to the growing body of research aimed at improving the accuracy and reliability of peak forecasting in time series data.

We adopt in this work the same approach as in **?** . We assume that the peak region is equal to $[T, h]$, where T represents the current date and h a specific time horizon. To model this hypothesis effectively, we consider our time series over the interval $[1, T]$ as a composite

of a Gaussian trend and noise. Our primary objective is to forecast the amplitude A, location L, and width S of the Gaussian trend based on observed data from the interval $[1, T]$. We introduce a criterion based on the weighted quantile technique, which enables us to determine the optimal number of observations required for accurately predicting the location L of the peak. Additionally, we conduct an analysis to assess the impact of the signal-to-noise ratio on our predictive accuracy. We show that in case of low signal-to-noise ratios, prediction accuracy diminishes. To address this challenge, we propose as in **?** a novel concept termed the "peak region entry and exit times".

The aim of our study is to extend the work of **?** to the quantile regression. First, we introduce our approach to selecting the minimization of the quantile regression and demonstrate, using simulated data, the number of observations required to accurately predict the location of the peak. This approach is based on the concept of the weighted quantile. Secondly, we establish that for high signal-to-noise ratios, the peak location prediction is correct. However, in cases of low `SNR` (indicating high noise levels), the accuracy of the prediction deteriorates. In these cases, we propose a new concept called "peak region entry time" and "peak region exit time". We illustrate this concept by parameterizing the entry time in the peak region and the exit time from the peak region by the tau-quantile when the data are the daily COVID-19 infections during the first wave in China **?**, France and Germany.

The remainder of this paper is organized as follows. In Section 2 we present our proposed model. More precisely, we consider Gaussian trend with additive generalized Laplace noise. Section **??** is dedicated to application results to forecast the peak incidence of daily COVID-19 infections during the initial wave in China, France, and Germany.

2 Quantile-Based Estimation Methodology

We consider the problem of estimating the amplitude, location, and spread of a unimodal signal corrupted by noise. The model is:

$$I(t) = A \exp\left(-\frac{(t-L)^2}{S^2}\right) + e(t), \quad t = 1, \ldots, T,$$

where $A > 0$ is the amplitude, $L > 0$ the peak location, $S > 0$ the spread (scale), and $e(t)$ is additive noise. Given observations $(I(t))_{t=1}^{T}$, our goal is to estimate the underlying parameters using quantile regression, i.e., minimizing the asymmetric loss

$$\rho_\tau(u) = u \cdot (\tau - \mathbf{1}_{\{u<0\}}),$$

for a fixed quantile level $\tau \in (0, 1)$.

We define the quantile objective:

$$f_\tau(T, a, l, s) = \frac{1}{T} \sum_{t=1}^{T} \rho_\tau \left(I(t) - a \exp\left(-\frac{(t-l)^2}{s^2} \right) \right).$$

For fixed candidate values of (l, s), the optimal amplitude a minimizes $a \mapsto f_\tau(T, a, l, s)$. This minimizer is a weighted τ-quantile of a transformed version of the data, as described below.

2.1 Weighted Quantile Estimation of Amplitude

Let $w(t) = \exp\left(-\frac{(t-l)^2}{s^2}\right)$ be the weight associated with time point t, and define:

$$x(t) = I(t) \exp\left(\frac{(t-l)^2}{s^2} \right).$$

Then the optimal amplitude a is the weighted τ-quantile of $x(t)$ with weights $w(t)$, i.e.,

$$a = \arg\min_{a'} \sum_{t=1}^{T} w(t) \rho_\tau(a' - x(t)).$$

The proof of the following Proposition is the same as in the work of **?**.

Proposition 1. *Let $(x(t), w(t))$ be a sequence of real values with positive weights $w(t) > 0$. Then, for any $\tau \in (0, 1)$, the weighted τ-quantile minimizes the weighted quantile loss:*

$$q_\tau = \arg\min_{a \in \mathbb{R}} \sum_{t=1}^{T} w(t)\, \rho_\tau(a - x(t)),$$

where $\rho_\tau(u) = u(\tau - \mathbf{1}\{u < 0\})$ is the quantile check function.

To compute q_τ, sort the values $x(t)$ in increasing order, obtaining a permutation π such that

$$x(\pi(1)) \leq x(\pi(2)) \leq \cdots \leq x(\pi(T)).$$

Define the reordered weights $w_\pi(t) = w(\pi(t))$. Let

$$k = \min \left\{ j \in \{1, \ldots, T\} : \sum_{t=1}^{j} w_\pi(t) \geq \tau \sum_{t=1}^{T} w(t) \right\}.$$

Then the weighted τ-quantile satisfies $q_\tau = x(\pi(k))$.

2.2 Quantile-Based Estimation Procedure

We define the following steps to jointly estimate the location $\hat{l}(T)$, amplitude $\hat{a}(T)$, and spread $\hat{s}(T)$:

Algorithm 1 Quantile-based estimation of peak parameters

Require: Time horizon T, upper bound $h > T$, quantile level $\tau \in (0, 1)$

1: For each candidate $l \in [1, h]$, find $s^*(l) = \arg\min_s f_\tau(T, a^*(l, s), l, s)$
2: Set $\hat{l}(T) = \arg\min_l f_\tau(T, a^*(l, s^*(l)), l, s^*(l))$
3: Set $\hat{s}(T) = s^*(\hat{l}(T))$
4: Set $\hat{a}(T) = a^*(\hat{l}(T), \hat{s}(T))$

Ensure: Estimated parameters $(\hat{l}(T), \hat{a}(T), \hat{s}(T))$

2.3 Remark on Asymmetry and Robustness

This method generalizes the classic Least Absolute Deviations (LAD) estimator, recovered at $\tau = 0.5$. By adjusting τ, we introduce asymmetry into the estimation loss, enabling the model to better adapt to skewed noise distributions. This makes the approach especially valuable in early-warning systems and epidemic peak detection where overestimation and underestimation have asymmetric consequences.

3 Quantile-Based Objective Function under Noise-Free Conditions

We first consider the idealized scenario where the signal is not corrupted by noise. The model of interest is given by

$$I(t) = A \exp\left(-\frac{(t-L)^2}{S^2}\right), \quad t = 1, \ldots, T,$$

where $A = 15000$, $L = 22$, $S = 10$, and $T \in \{2, 3, 10, 20, 21\}$. The quantile-based estimation procedure is implemented by minimizing the following objective function:

$$f_\tau(T, a, l, s) = \frac{1}{T} \sum_{t=1}^{T} \rho_\tau \left(I(t) - a \exp\left(-\frac{(t-l)^2}{s^2}\right)\right),$$

where $\rho_\tau(u) = u(\tau - \mathbf{1}_{\{u<0\}})$ denotes the asymmetric quantile loss. The amplitude a is computed via a weighted τ-quantile, as described in Section 2.

Since the goal is to predict the location L of the peak before it becomes observable, we restrict the minimization over candidate peak locations $l > T$. For each value of $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, we compute and visualize the objective curve $l \mapsto f_\tau(T, a(T, l), l, s(T, l))$ for the selected values of T (see Figure 1).

Remark. For each fixed sample size T, we computed the value of the minimized objective function at the estimated parameters $(\hat{a}, \hat{l}, \hat{s})$. The results indicate that the lowest objective value is typically achieved for $\tau = 0.95$, suggesting that upper quantiles align particularly well with the noiseless signal structure. The only exception is at $T = 20$, where $\tau = 0.75$ yields a slightly smaller loss. This empirical observation may help guide quantile selection when prior

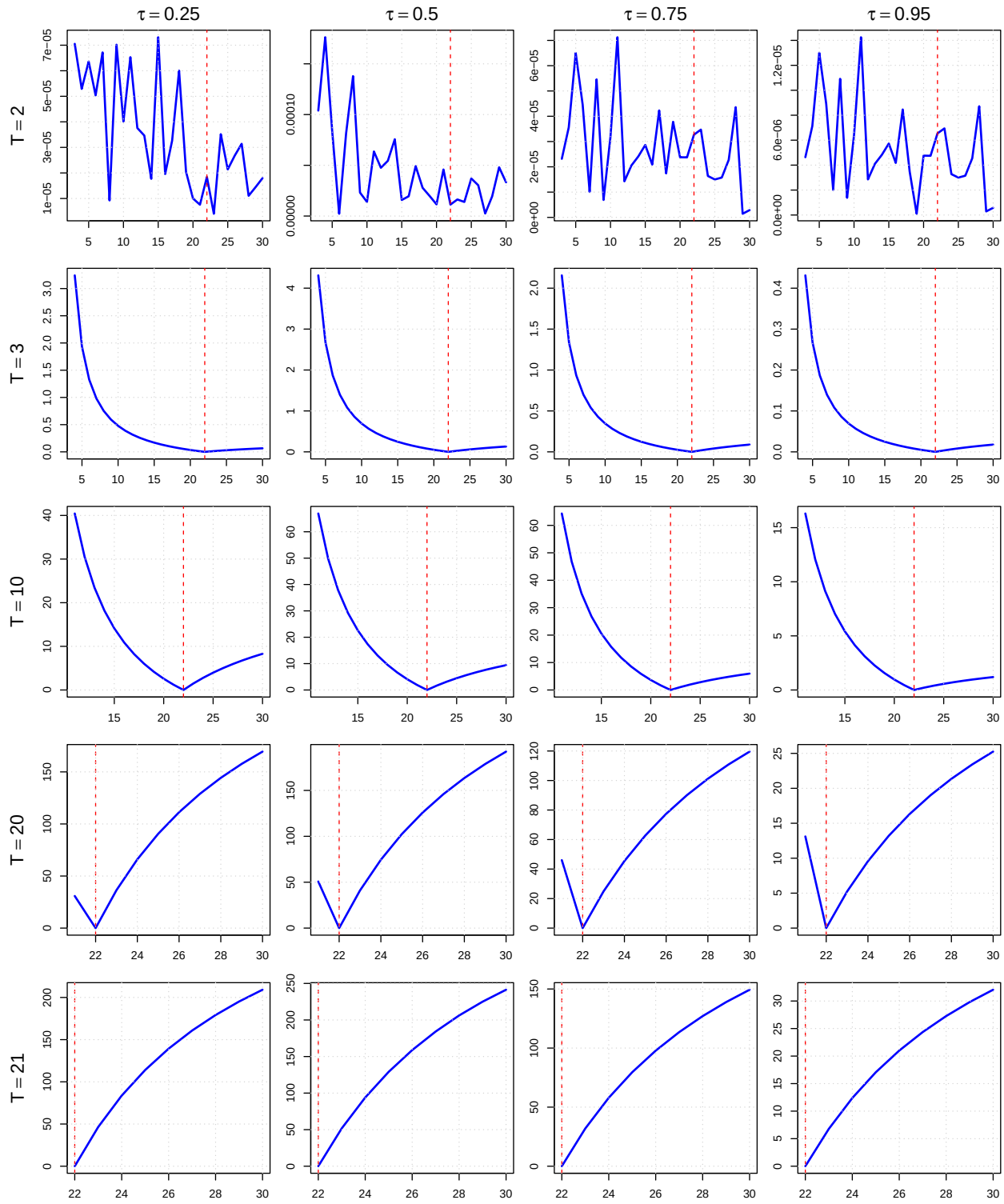


Figure 1: Objective function $l \mapsto f_\tau(T, a, l, s)$ computed for $\tau \in \{0.25, 0.5, 0.75, 0.95\}$ and $T \in \{2, 3, 10, 20, 21\}$, under a noise-free signal. The vertical dashed line indicates the true peak location $L = 22$, and the search is restricted to values $l > T$.

knowledge of signal shape or noise properties is limited.

4 Quantile-Based Estimation with τ-Aligned Noise

To analyze how quantile-aligned noise affects peak estimation, we introduce a unified noise model based on the *general asymmetric Laplace* (GAL) distribution. This flexible family allows both symmetric and asymmetric noise structures while maintaining the property that the τ-quantile of the noise is at zero.

4.1 Asymmetric Laplace Noise Aligned with the τ-Quantile

We consider an asymmetric Laplace distribution parametrized by a location parameter $m \in \mathbb{R}$, a scale parameter $\lambda > 0$, and an asymmetry parameter $k > 0$. Its density is given by

$$f_{m,\lambda,k}(x) = \frac{\lambda}{k + k^{-1}} \begin{cases} \exp\left(\dfrac{\lambda(x-m)}{k}\right), & x < m, \\ \exp\left(-\lambda k(x-m)\right), & x \geq m. \end{cases}$$

The parameter m is the mode of the distribution, while k controls its asymmetry and λ controls its concentration. The distribution function evaluated at the mode is

$$\mathbb{P}(X < m) = \frac{\lambda}{k + k^{-1}} \int_{-\infty}^{m} \exp\left(\frac{\lambda(x-m)}{k}\right) dx = \frac{k^2}{1+k^2}.$$

Therefore, to align the asymmetric Laplace noise with the τ-quantile regression criterion, we require $\mathbb{P}(e < 0) = \tau$, and $\tau \in (0, 1)$. Since the location of the noise can be fixed at zero, we set $m = 0$. The corresponding asymmetry parameter is then determined by

$$\frac{k^2}{1+k^2} = \tau,$$

or equivalently,

$$k = \sqrt{\frac{\tau}{1-\tau}}.$$

Thus, the τ-aligned asymmetric Laplace noise has density

$$f_{\tau,\lambda}(x) = \frac{\lambda}{k + k^{-1}} \begin{cases} \exp\left(\dfrac{\lambda x}{k}\right), & x < 0, \\ \exp(-\lambda k x), & x \geq 0, \end{cases} \qquad k = \sqrt{\frac{\tau}{1-\tau}}.$$

By construction, $\mathbb{P}(e < 0) = \tau$. In particular, when $\tau = 1/2$, we have $k = 1$, and the asymmetric Laplace distribution reduces to the symmetric Laplace distribution centered at zero. Hence, the symmetric Laplace case is recovered as a special case of the τ-aligned asymmetric Laplace model.

4.2 Shifted Symmetric Laplace (Aligned at τ)

We now consider a heuristic construction in which the additive noise is drawn from a symmetric Laplace distribution and then shifted so that its τ-quantile is exactly zero. This ensures that the noise is statistically aligned with the asymmetric loss ρ_τ, i.e., $\mathbb{P}(e(t) < 0) = \tau$, even though the original noise is symmetric around zero. More precisely, let $E(t) \sim \text{Laplace}(0, b)$, and define $e(t) = E(t) - q_\tau$, where q_τ is the theoretical τ-quantile of the Laplace$(0, b)$ distribution:

$$q_\tau = \begin{cases} b \log(2\tau), & \text{if } \tau < 0.5, \\ -b \log(2(1-\tau)), & \text{if } \tau \geq 0.5. \end{cases}$$

By construction,

$$\mathbb{P}(e(t) < 0) = \mathbb{P}(E(t) < q_\tau) = \tau.$$

Thus, the τ-quantile of the shifted noise distribution is zero. Importantly, the shift does not affect the variance. Since $\text{Var}(E(t)) = 2b^2$, we also have $\text{Var}(e(t)) = 2b^2$. Consequently, if the signal-to-noise ratio is defined by

$$\text{SNR} = \frac{A}{\sqrt{\text{Var}(e(t))}},$$

the Laplace scale parameter is chosen as

$$b = \frac{A}{\sqrt{2}\ \text{SNR}}.$$

We simulate observations according to

$$I(t) = A \exp\left(-\frac{(t-L)^2}{S^2}\right) + e(t),$$

where $A = 1$, $L = 22$, $S = 10$, and $\text{SNR} = 1000$. For each $\tau \in \{0.25, 0.5, 0.75, 0.95\}$ and $T \in \{4, 6, 8, 9, 10\}$, we evaluate the objective function

$$f_\tau(T, l) := \min_s \frac{1}{T} \sum_{t=1}^{T} \rho_\tau \left(I(t) - a(T, l, s) \exp\left(-\frac{(t-l)^2}{s^2}\right)\right),$$

where $a(T, l, s)$ is computed as the weighted τ-quantile of the transformed observations

$$x(t) = I(t) \exp\left(\frac{(t-l)^2}{s^2}\right),$$

with weights

$$w(t) = \exp\left(-\frac{(t-l)^2}{s^2}\right).$$

Figure 2 shows the resulting profiles $l \mapsto f_\tau(T, l)$ across different quantile levels and sample sizes. The red dashed line marks the true peak location $L = 22$.

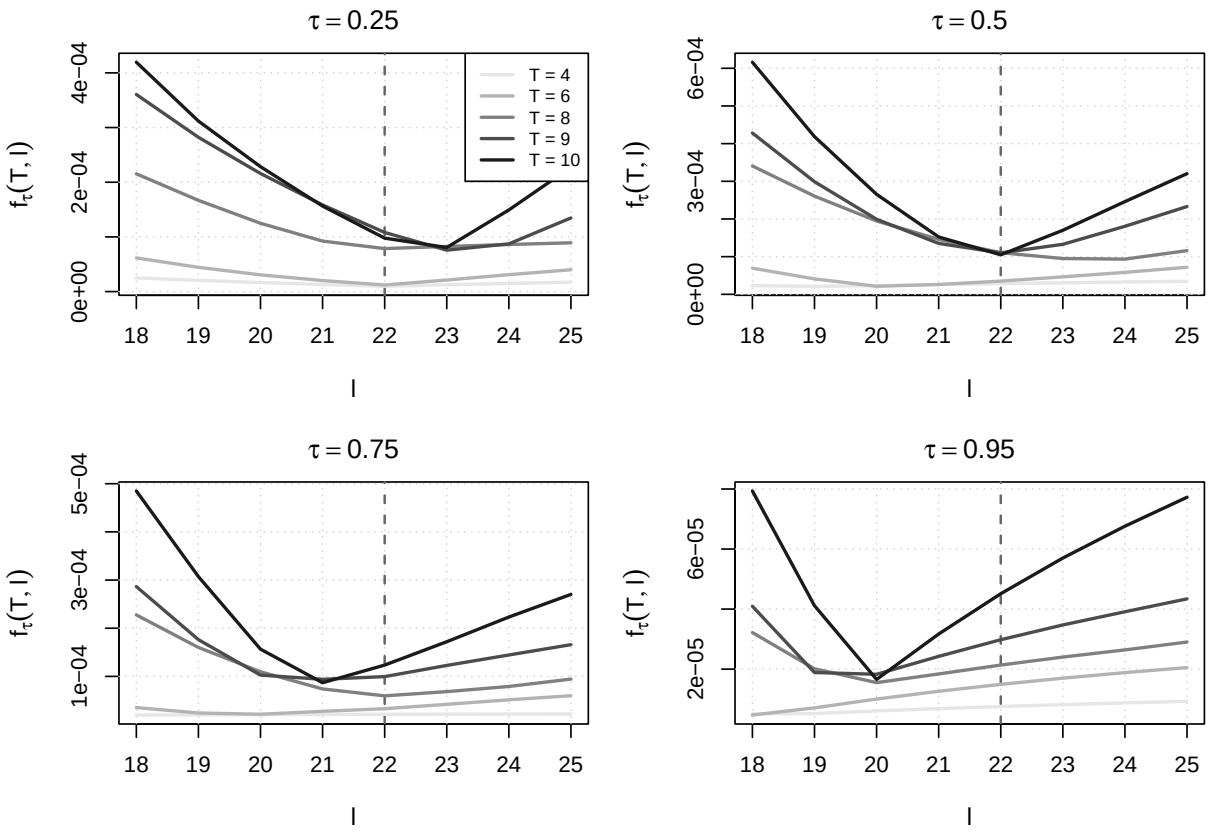


Figure 2: Objective profiles $l \mapsto f_\tau(T, l)$ under shifted symmetric Laplace noise for quantile levels $\tau \in \{0.25, 0.5, 0.75, 0.95\}$ and time horizons $T \in \{4, 6, 8, 9, 10\}$. Each panel corresponds to a single quantile level τ. Within each panel, curves correspond to the different horizons T. The dashed vertical line indicates the true peak location $L = 22$.